

- 9 (a) A neuromuscular junction allows the transmission of an action potential from a motor neurone to a striated muscle fibre, causing it to contract.

Fig. 9.1 is a graph of an action potential in a motor neurone.

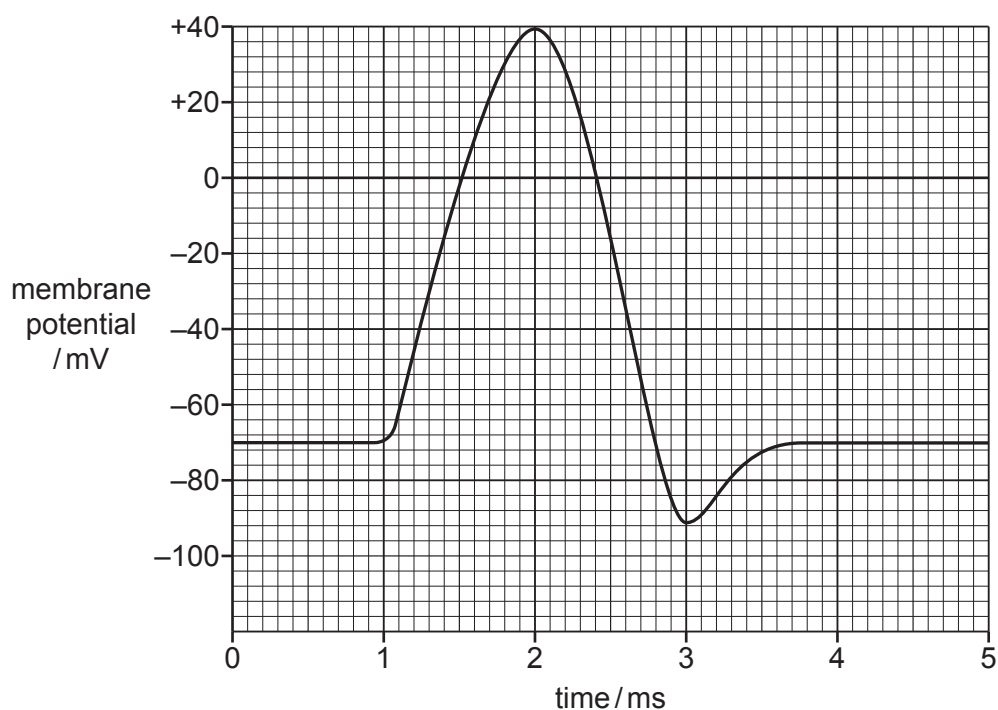


Fig. 9.1

Fig. 9.2 is a graph of an action potential in a striated muscle fibre.

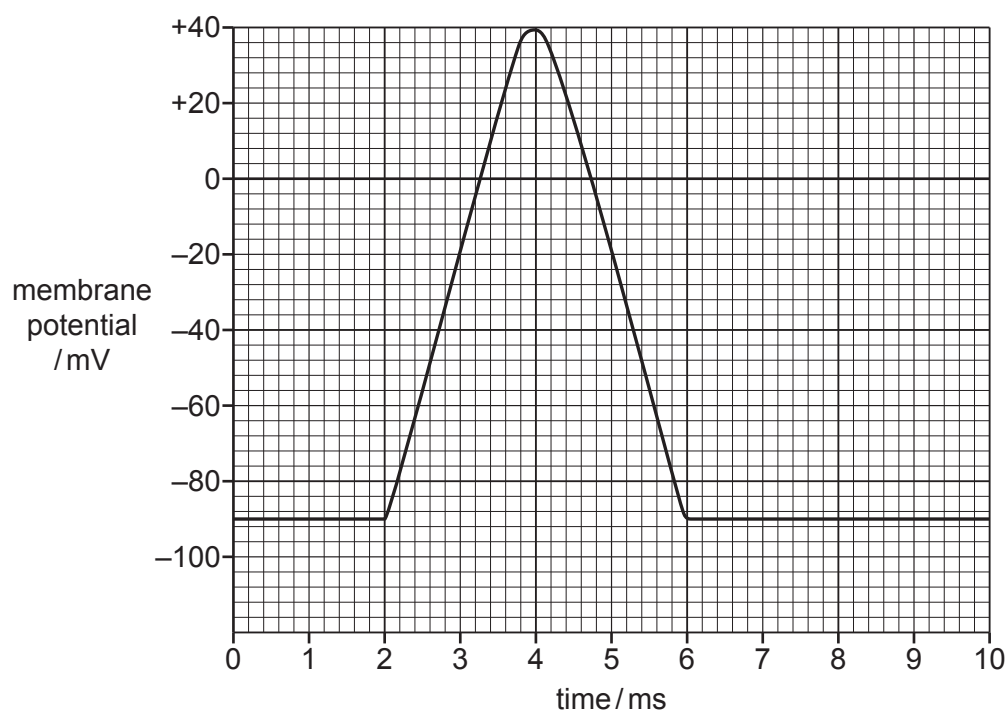


Fig. 9.2

[4]

- (b) There are three phases in the contraction of a striated muscle: latent phase, contraction phase and relaxation phase.

The tension in a muscle represents the degree of contraction of its fibres.

Fig. 9.3 is a graph of the tension in a striated muscle during the three phases of contraction.

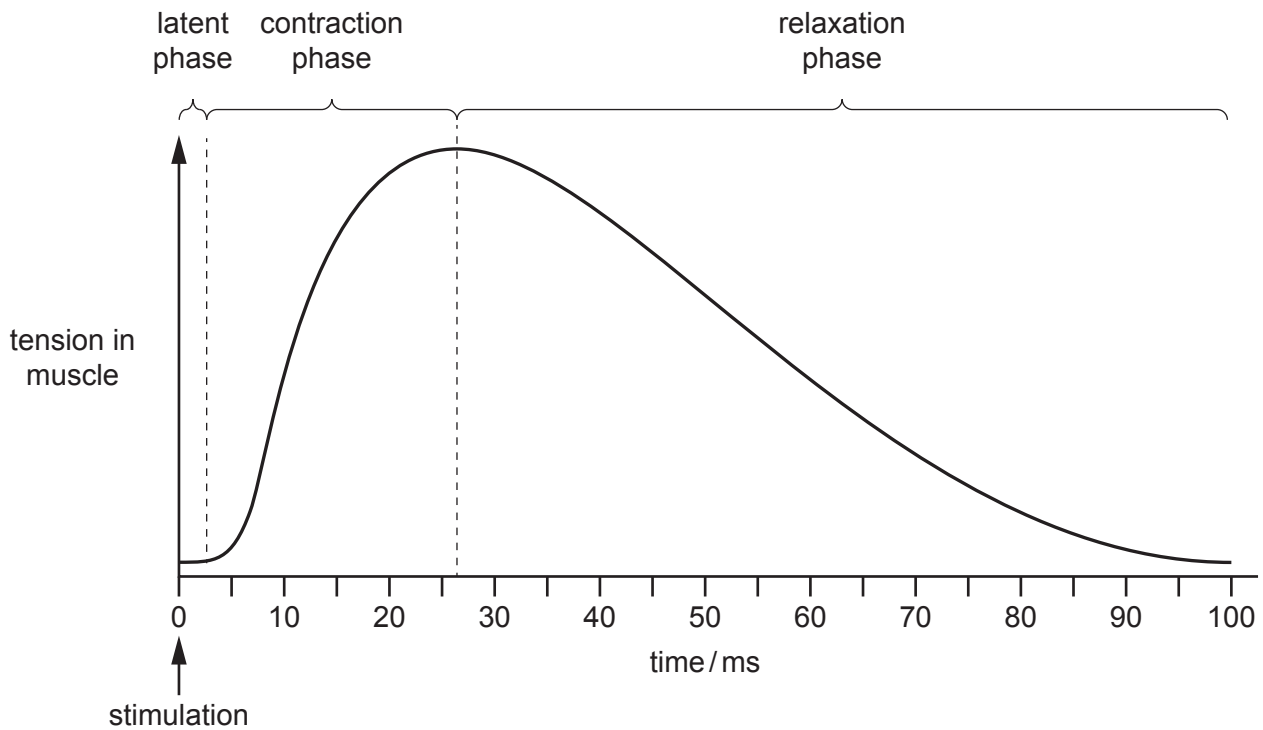


Fig. 9.3

- (i) With reference to Fig. 9.3, explain what is happening in the striated muscle fibre during the latent phase.

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- (ii) Suggest why the relaxation phase shows a gradual decrease in muscle tension.

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[Total: 8]